ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers.

SECTION A

QUESTION 1

1.1 D ✓
1.2 B ✓
1.3 D ✓
1.4 B ✓
1.5 C ✓
1.6 A ✓
1.7 B ✓
1.8 D ✓
1.9 A ✓
1.10 C ✓

(10 × 1) [10]

QUESTION 2

2.1 True ✓
2.2 False ✓
2.3 True ✓
2.4 True ✓
2.5 False ✓
2.6 True ✓
2.7 False ✓
2.8 True ✓
2.9 True ✓
2.10 False ✓

(10 × 1) [10]

QUESTION 3

3.1 F ✓
3.2 E ✓
3.3 J ✓
3.4 D ✓
3.5 G ✓
3.6 C ✓

3.7 I ✓

3.8 H ✓

3.9 B ✓

3.10 A ✓

(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Any FOUR: always put the hot iron back in its stand ✓; work in a well-ventilated area / use a fume extractor ✓; wear safety glasses ✓; never touch the tip ✓; switch off and unplug after use; wash hands after handling solder; keep the cable away from the hot tip (4)

4.2 Current – the flow of electric charge (electrons) ✓, ampere (A) ✓; voltage – the potential difference / electrical pressure that pushes current ✓, volt (V) ✓; resistance – opposition to the flow of current ✓, ohm (Ω) ✓ (6)

4.3 The current through a conductor is directly proportional to the voltage across it at constant temperature ✓; V = I × R ✓ (2)

4.4 $V_R = 5 - 2 = 3\ V$ ✓; $R = V / I$ ✓ $= 3 / 0{,}02$ ✓ $= 150\ \Omega$ ✓ (4)

4.5 (a) $P = V \times I = 12 \times 1{,}5$ ✓ $= 18\ W$ ✓ (b) $R = V / I = 12 / 1{,}5$ ✓ $= 8\ \Omega$ ✓ (4)

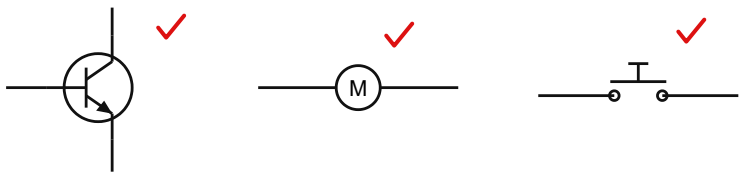
[20]

QUESTION 5

5.1 (a) $R_T = 100 + 220 + 330$ ✓ $= 650\ \Omega$ ✓ (b) $I = 9 / 650$ ✓ $= 0{,}014\ A$ (13,85 mA) ✓ (c) $V = 0{,}01385 \times 220$ ✓ $= 3{,}05\ V$ ✓ (6)

5.2 $R_T = (60 \times 30) / (60 + 30)$ ✓ $= 1\,800 / 90$ ✓ $= 20\ \Omega$ ✓ (3)

5.3



(a) NPN transistor (b) DC motor (c) push button (NO)

((a) NPN transistor with the emitter arrow pointing out 1; (b) motor – circle with M 1; (c) push button, normally open 1)

(3)

5.4 When the transistor switches off, the coil produces a high reverse voltage (back EMF) ✓; the diode (flyback diode) gives this current a safe path ✓ and protects the transistor / microcontroller from damage ✓. (3)

[15]

QUESTION 6

6.1 (a) $45 = 32 + 8 + 4 + 1$ ✓ → 101101 ✓ (b) $128 + 64 + 32 + 4 + 1$ ✓ $= 229$ ✓ (c) $1010 = \text{A}, 1111 = \text{F}$ ✓ → AF ✓ (6)

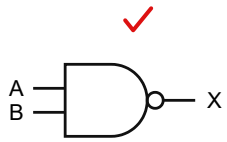
6.2 $107 + 29 = 136$ ✓ → 1000 1000 ✓ (2)

6.3 $2^8 - 1$ ✓ $= 255$ ✓ $(1111\ 1111)$ (2)

[10]

QUESTION 7

7.1



A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

(Symbol: AND shape with a small circle (bubble) at the output 1; truth table 4)

(5)

7.2.1 Gate 1: AND ✓; gate 2: NOT (inverter) ✓; gate 3: OR ✓ (3)

7.2.2 $\text{X} = \text{A}\cdot\text{B} + \bar{\text{C}}$ ✓✓ $(\text{X} = (\text{A AND B}) \text{ OR } (\text{NOT C}))$ (2)

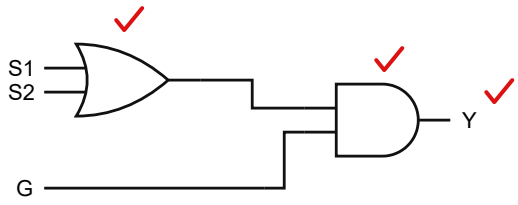
7.2.3 Truth table (1 mark per correct row):

A	B	C	D = A·B	E = C̄	X
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	1	0	1

(8)

7.2.4 No ✓ – X = 0, because only one bump sensor is pressed (D = 0) and the safety key is in place (E = 0) ✓. (2)

7.3



(Y = G · (S1 + S2) 2. Circuit: S1 and S2 into an OR gate 1; OR output and G into an AND gate 1; output labelled Y 1)

(5)

[25]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE